

4th Progress Report

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Outcome of the previous meeting

- Reviewed almost ready draft of research questions.
- Discussed more possible research directions:
 1. RL gain tuning (meta-gain tuning)
 2. Multistep IDHP, research could be to investigate this algorithm on its' own or to incorporate it with "offline" DRL methods (SAC, DSAC, TD3, ...)

Description of the results since last meeting

- Derived the multi-step and eligibility trace extensions to IDHP
- The *almost* final research objective and questions are stated as:

Research Objective:

To advance the state-of-the-art reinforcement learning based flight controllers and further the technological readiness level of the Flying-V, by developing an intelligent and fault tolerant flight control system for the Flying-V.

Research Questions:

1. What reinforcement learning algorithm can yield a flight controller that is the most tolerant to faults?
 - (a) What RL algorithms are considered to be state-of-the-art?
 - (b) In what ways can fault tolerance be built into an RL-based flight controller?
 - (c) Which algorithms demonstrate the best fault tolerance?
 - (d) How are the reference tracking performance of these algorithms?
2. What are the flight control-related challenges when designing a controller for the Flying-V?
 - (a) What are potential fault scenarios that warrant attention from the AFCS system in the Flying-V?
 - (b) What criteria should be used to characterize a controller's fault tolerance (fault tolerance success rate, nMAE of tracking error, variance of nMAE...)?
3. How can the identified algorithm be applied to control the Flying-V?
 - (a) How should the flight control system be structured (cascaded control structure, what signals to feedback)?

- (b) What is the MDP in the case of controlling the Flying-V (the state variables, action variables, and environment to be controlled)?
- 4. How does the implemented flight controller perform during nominal flight and in the presence of faults?
 - (a) Noting the possible fault scenarios and handling qualities of the Flying-V, what flight scenarios should be designed to test the performance of the controller (what faults to use, what tracking signals to use, superimposed faults...)?
 - (b) What is the degree of fault tolerance of the implemented controller (I added this to have some kind of absolute judgment of the performance of the controller, but will not do something like impose a requirement that must be satisfied by the controller)?
 - (c) How well does the nominal flight performance of the proposed flight controller compare to other state-of-the-art controllers?
 - (d) How are the fault tolerance characteristics of the proposed flight controller compared to other state-of-the-art controllers?
- In light of this objective, and of the literature study performed thus far, the following is surmised:

The online-offline hybrid approach to applying RL to flight controllers seems the most promising solution to tackle the challenge of fault tolerance. The “offline” algorithms come in the form of model-free actor-critic methods, which while being relatively sample inefficient to the “online” algorithms, are firstly able to generalize to more control tasks as they learn policies and value function estimates by repeated traversals across the state-action space, and secondly can provide an initialization condition for the “online” algorithms which have been pre-trained and tested rather than from tabular rasa, otherwise requiring online system identification and more noticeable & inconvenient persistent excitations.

- A short outline of the research plans is as follows:

The final goal of the research project is implementing and evaluating an online-offline hybrid controller for attitude control of the Flying-V. At this stage, it is difficult to tell whether this goal is too achievable or overly ambitious, but to be conservative keeping in mind time constraints it will be assumed that this goal is ambitious. Therefore, contingencies will be planned in the form of placing intermediate sub-goals between commencing the first research phase and achieving the final goal:

1. To implement an augmented IDHP algorithm, this implementation is prioritized as this incremental model-based algorithm is considered to be highly adaptive and can thus provide truly online control, which will have better odds of controlling the aircraft in the face of sudden changes in system dynamics as a result of faults.
2. To implement the “offline” learning algorithms, which can take the form of as basic as SAC, DSAC, or even the state-of-the-art RUN-DSAC algorithm
3. To merge the two implementations, using the offline trained critic networks to initialize the online ADP algorithm.

Achieving the first sub-goal would yield the academic contribution of this MSc research, by contributing towards the development of the IDHP algorithm.

Achieving the second sub-goal will yield the societal contribution of this research, as an offline-online hybrid algorithm has improved flight performance and fault tolerance characteristics than a purely online or offline-trained controller.

Plans for next 2 weeks

1. Write literature study and read the final few papers I have left.
2. Write research plan, implementations that need to be done, analyses that need to be done, order to do the tasks, and approximate duration of tasks.